

AYYOUB LAMSSAGGAD

Backend Software Engineer

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PROFESSIONAL EXPERIENCE

- Backend Software Engineer**
Matious Digital with LUBU Technologies
Mar 2025 – Present Morocco (Hybrid)
 - Designed and delivered a modular back-end architecture (FastAPI, PostgreSQL) for the LUBU platform, enabling scalable management of accounts, athletes, smart footwear devices, and training sessions.
 - Built advanced biomechanical analysis tools leveraging Inertial Measurement Unit (IMU) sensor data to convert complex movement patterns into practical, actionable insights for coaches, athletes, and medical staff.
 - Engineered AES-based encryption/decryption middleware to transparently secure sensitive request and response data, implementing robust protection measures in compliance with HIPAA regulations.
 - Integrated Stripe payment workflows and webhook handling to automate subscription billing, verify transactions, and synchronize order data with the backend, ensuring secure and reliable e-commerce operations.
- Backend Software Engineer**
Veylinx
Nov 2022 – Mar 2025 Netherlands (Remote)
 - Built and maintained scalable RESTful APIs for the Veylinx market research platform using Django and FastAPI, ensuring system performance as more experiments and users were onboarded to the platform.
 - Designed a modular reporting system integrating Django ORM with OpenSearch, enabling business and research users to perform dynamic analysis on behavioral experiment data.
 - Boosted query performance through strategic indexing and query optimization, directly contributing to faster experimentation cycles and more actionable insights for clients.
 - Developed hundreds of automated tests to ensure the Veylinx platform's stability and performance as new features were introduced.
- Ph.D. Researcher in Computer Science**
Moulay Ismail University
2020 - 2022 Morocco
 - Conducted advanced research on Intelligent Transportation Systems (ITS), with a focus on securing data integrity and detecting threats.
 - Collaborated with academic and industry professionals, publishing key findings in IEEE Access, a highly reputable journal.

EDUCATION

- Master's Degree in Information Systems Security.**
Ibn Tofail University
2017 - 2019 Kenitra , Morocco



ABOUT ME

Backend software engineer with hands-on experience designing scalable systems. I'm particularly interested in the intersection of AI and cloud/edge computing, as well as how backend architecture can support intelligent, distributed systems. My work focuses on building modular infrastructures and optimizing data flows for high-performance applications. Now motivated to deepen my expertise through research, I aim to explore how AI can be effectively integrated into cloud environments. I bring both practical experience and a strong desire for academic growth to advance and drive research forward.

TECHNICAL SKILLS

- Programming Languages:** Python, Java
- Frameworks:** Django, FastAPI
- Databases:** PostgreSQL, Redis, OpenSearch
- DevOps:** Docker, GitHub Actions
- Cloud:** AWS (EC2, S3, RDS)
- Monitoring:** Datadog, Sentry, CloudWatch
- Version Control:** Git, GitHub
- Other Skills:** Microservices, RESTful API

SOFT SKILLS

- Critical Thinking, Attention to Detail, Problem-Solving, Strategic Planning

LANGUAGES

- Arabic : Native
- English : Professional Proficiency
- French : Limited Proficiency

PUBLICATIONS

- A Survey on the Current Security Landscape of Intelligent Transportation Systems.
IEEE Access
2021 DOI: 10.1109/ACCESS.2021.3050038